

Notes: Froot and Stein (QJE, 1991)

Exchange Rates and Foreign Direct Investment: An Imperfect Capital Markets Approach

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1 Big picture

1.1 Question

Froot and Stein ask why foreign direct investment into the United States rose sharply when the dollar depreciated. In a frictionless global capital market, exchange rates should not alter who can buy domestic productive assets: domestic and foreign investors should face the same risk-adjusted cost of capital. The paper challenges that frictionless benchmark by allowing capital markets to be globally integrated but informationally imperfect.

The core question is: can a currency depreciation shift control of domestic assets toward foreign buyers by changing relative wealth, even when foreign and domestic investors can borrow in the same world capital market?

1.2 Main answer

Yes. If external finance is more costly than internal finance, wealth matters for bidding on information-intensive assets. A domestic currency depreciation raises the domestic-currency value of wealth held by foreign investors. This lowers their effective financing constraint relative to domestic investors and lets them bid more aggressively for direct-control assets.

The key is not that foreign investors face a lower world interest rate. They do not. The key is that direct investments require internal funds because outsiders cannot perfectly verify project payoffs. A depreciation therefore shifts relative internal wealth and changes who wins asset auctions.

1.3 Core finding

The empirical evidence supports a distinctive prediction of the model. Foreign direct investment inflows are negatively related to the value of the dollar, whereas passive portfolio inflows are not. This difference is important because it separates the paper's mechanism from stories in which all capital inflows mechanically rise when the dollar falls.

1.4 Economic takeaway

The paper provides a formal version of the intuitive “cheap dollar” story, but with a precise friction: costly external finance. A weak domestic currency can make foreign investors relatively wealthy and therefore more able to acquire control of domestic assets. The mechanism is strongest for assets that are information-intensive and weakest for passive securities such as Treasury bonds or diversified portfolios.

2 Incremental contribution of the paper

2.1 Relative to standard open-economy capital-market theory

The standard perfect-capital-mobility view implies that exchange rates should not affect the composition of capital inflows. The current account pins down net capital inflow, but not whether the inflow is Treasury purchases, portfolio equity, real estate, or FDI. Froot and Stein add imperfect

external finance and show that the exchange rate can affect the composition of inflows, especially the share going into direct investment.

2.2 Relative to industrial-organization theories of FDI

Classical FDI theory emphasizes control advantages: foreign firms acquire domestic assets because they can operate them more efficiently. Froot and Stein do not deny this channel, but add a financing channel. Even if domestic and foreign entrepreneurs have the same operating ability, exchange-rate movements can affect who controls the asset by shifting relative wealth.

2.3 Relative to empirical FDI work

The paper does more than show that FDI is correlated with the dollar. It tests a sharp implication: FDI should behave differently from passive capital inflows because direct investment is more information-intensive. The empirical tables show that the dollar-FDI relation appears in direct investment, industries, and many transaction types, while portfolio inflows behave differently.

2.4 Relative to welfare discussions

The paper cautions against a simple protectionist conclusion. A depreciation-induced foreign acquisition can, in a special case, reduce domestic welfare if domestic entrepreneurs are severely credit constrained and assets sell for less than their value under domestic control. But this is not a general case. Competition among foreign bidders or new capital formation can overturn the pessimistic welfare implication.

3 Model setup

3.1 Agents and assets

There are two periods and risk-neutral agents. Agents can invest in riskless domestic and foreign one-period bonds, and can also bid for risky indivisible direct-investment assets in the domestic country. The domestic riskless bond pays interest rate r in domestic currency. The foreign riskless bond pays r^* in foreign currency.

The exchange rate e is the domestic-currency price of foreign currency. Under risk neutrality and perfect capital mobility for riskless bonds, uncovered interest parity is

$$r - r^* = \frac{E_1(e_2) - e_1}{e_1}.$$

The paper normalizes $E_1(e_2) = 1$ and $r^* = 0$, so

$$1 + r = \frac{1}{e}.$$

A domestic depreciation is an increase in e , which raises the domestic-currency value of foreign-currency wealth.

3.2 Direct investment technology

Each direct asset can be run either by a domestic entrepreneur or by a foreign entrepreneur. The payoff under either operator is uniformly distributed on $[0, X]$. Thus expected gross payoff is $X/2$. The simplification that both investors have the same operating ability isolates the financing channel.

The payoff is observed costlessly by the owner-manager, but an outside creditor must pay a monitoring cost c to verify it. This creates costly external finance. The monitoring cost is the key friction: if $c = 0$, direct investment can be externally financed without loss, and wealth no longer matters.

4 External finance and loan supply

4.1 Debt contract

With costly state verification, the optimal contract is debt. The entrepreneur borrows L and promises repayment D . If realized payoff $x \geq D$, the lender receives D . If $x < D$, the lender monitors, pays cost c , and receives $x - c$.

The lender's expected payoff is

$$R_D^L = \int_0^D \frac{x - c}{X} dx + \int_D^X \frac{D}{X} dx = D - \frac{D^2}{2X} - \frac{cD}{X}.$$

Competitive lenders require

$$R_D^L = (1 + r)L.$$

Thus the loan supply generated by a promised repayment D is

$$L(D) = \frac{1}{1 + r} \left(D - \frac{D^2}{2X} - \frac{cD}{X} \right).$$

4.2 Maximum borrowing

To find the maximum loan, differentiate $L(D)$ with respect to D :

$$\frac{dL}{dD} = \frac{1}{1 + r} \left(1 - \frac{D}{X} - \frac{c}{X} \right).$$

The maximizing repayment is

$$D^* = X - c.$$

Substitution gives

$$L^{\max} = \frac{(X - c)^2}{2X(1 + r)}.$$

When $c = 0$, this becomes $X/[2(1 + r)]$, exactly the present value of expected payoff. When $c > 0$, the maximum loan is strictly below the asset's expected present value, so the entrepreneur may be credit rationed.

5 Entrepreneurial reservation price

5.1 Entrepreneur payoff

If the entrepreneur promises repayment D , his expected residual payoff is

$$R_D^E = \int_D^X \frac{x - D}{X} dx = \frac{D^2}{2X} - D + \frac{X}{2} = \frac{(X - D)^2}{2X}.$$

If he uses wealth w in the investment and is not credit rationed, the marginal condition is

$$R_D^E = (1 + r)w.$$

Solving for the promised repayment gives

$$D = X - \sqrt{2X(1 + r)w}.$$

5.2 Reservation bid away from the credit-rationed corner

Adding the lender and entrepreneur payoffs gives total surplus net of monitoring costs:

$$R_D^L + R_D^E = \frac{X}{2} - \frac{cD}{X}.$$

Since the total purchase price is $P = L + w$,

$$(1 + r)P = \frac{X}{2} - \frac{cD}{X}.$$

Substituting the nonrationed repayment $D = X - \sqrt{2X(1 + r)w}$ yields

$$P(X, w, c) = \frac{1}{1 + r} \left[\frac{X}{2} - c \left(1 - \sqrt{\frac{2w(1 + r)}{X}} \right) \right].$$

This formula shows why wealth matters: greater internal wealth reduces reliance on costly outside finance and lets the entrepreneur bid more.

5.3 Three regions

The reservation price is piecewise:

$$P(X, w, c) = \begin{cases} \frac{(X - c)^2}{2X(1 + r)} + w, & 0 < w < \frac{c^2}{2X(1 + r)}, \\ \frac{1}{1 + r} \left[\frac{X}{2} - c \left(1 - \sqrt{\frac{2w(1 + r)}{X}} \right) \right], & \frac{c^2}{2X(1 + r)} < w < \frac{X}{2(1 + r)}, \\ \frac{X}{2(1 + r)}, & w > \frac{X}{2(1 + r)}. \end{cases}$$

The regions have clear meanings. In Region 1, the entrepreneur is fully credit rationed and can bid only the maximum loan plus wealth. In Region 2, internal wealth relaxes the external-finance problem but does not eliminate it. In Region 3, the entrepreneur can self-finance the full present value, so additional wealth does not raise the bid.

5.4 Comparative statics

The proposition implies

$$\frac{\partial P}{\partial c} \leq 0, \quad \frac{\partial P}{\partial w} \geq 0, \quad \frac{\partial P}{\partial X} > 0.$$

Higher monitoring costs reduce reservation prices because external finance destroys more value. Higher wealth raises reservation prices because internal funds are more valuable than external funds. Higher operating ability X raises value under control.

When $c = 0$, the reservation price collapses to

$$P = \frac{X}{2(1+r)},$$

which is independent of wealth. This is the frictionless benchmark.

6 Exchange rates and foreign direct investment

6.1 Relative wealth channel

Let the domestic entrepreneur have domestic-currency wealth w^d . Let the foreign entrepreneur have foreign-currency wealth W^f . Its domestic-currency value is

$$w^f = eW^f.$$

A domestic depreciation means e rises. Therefore, the foreign entrepreneur's domestic-currency wealth rises relative to the domestic entrepreneur's wealth.

Because $P(X, w, c)$ is increasing in w when external finance is costly, the depreciation raises foreign reservation prices. More foreign entrepreneurs win auctions for domestic direct-investment assets. This is the paper's main theoretical channel.

6.2 Role of information intensity

The exchange-rate sensitivity is stronger when c is higher. High c means direct control is more information-intensive and harder to finance externally. For low c , wealth matters less. For $c = 0$, wealth does not matter at all and exchange rates do not affect who buys the asset.

Thus the model predicts a ranking across assets:

direct, information-intensive assets: high sensitivity to exchange rates,

passive portfolio assets: low or no sensitivity to exchange rates.

6.3 Current account interpretation

The model can be embedded in a general-equilibrium setting. The paper gives a simple two-good case in which the two currencies are goods. A depreciation can raise period-1 FDI, while the current account remains balanced because offsetting portfolio flows finance the direct-investment inflow. This helps separate the composition of capital inflows from the net current account position.

7 Welfare implications

7.1 Why welfare can be ambiguous

The exchange-rate-FDI result is robust, but the welfare conclusion is not. The model contains a distortion: entrepreneurs may be credit constrained. In a second-best setting, moving ownership to the highest bidder need not maximize domestic welfare if the highest bidder wins because of financing capacity rather than productive superiority.

7.2 Welfare-reducing acquisition case

If the domestic entrepreneur is in Region 1, he may be unable to bid the full value of the asset under his control. A foreign entrepreneur whose wealth rises after depreciation can outbid him even if the asset is equally or more valuable in domestic hands. In that case, the asset can be sold at a price below its domestic-control value.

The paper's numerical case uses $X = 10$, $c = 3$, $w^d = 0.30$, and $W^f = 0.25$. Initially, with $e = 1$, the domestic entrepreneur can bid slightly more. After a depreciation to $e = 2$, foreign wealth in domestic currency doubles, and the foreign entrepreneur wins. The asset is effectively sold too cheaply from the domestic welfare perspective.

7.3 Why protectionism does not automatically follow

Several factors can reverse the negative welfare conclusion. If the domestic entrepreneur is in Region 2 or 3, he can bid closer to the value of the asset, so the foreign winner must pay at least the domestic value. If there are multiple foreign bidders, competition transfers more surplus to the domestic seller. If FDI creates new capital or technology spillovers, welfare can improve.

The paper's policy message is therefore balanced: exchange rates can affect control of domestic assets, but the welfare effect depends on auction structure, domestic financing constraints, foreign competition, and whether the FDI is an acquisition or new capital formation.

8 Alternative explanations considered

8.1 Tax changes

Tax reforms can change the relative valuation of U.S. assets for domestic and foreign investors. The 1981 and 1986 U.S. tax reforms could matter. The paper argues, however, that tax timing cannot easily explain why the exchange-rate-FDI relation appears before 1980 and remains in studies that control for tax effects.

8.2 All capital inflows move together

A mechanical alternative is that FDI is a stable share of gross capital inflows. If the dollar falls and all foreign inflows rise in dollar terms, FDI would rise too. The paper's empirical response is to compare FDI with official assets, Treasury securities, and portfolio stocks and bonds. The model predicts that FDI should behave differently.

8.3 Sticky asset prices

Sticky domestic asset prices could make assets look cheap to foreign buyers after depreciation. The paper notes that in perfect capital markets this opportunity would be equally available to domestic investors. Under imperfect capital markets, however, price sluggishness becomes part of the paper's own mechanism: domestic bidders cannot fully raise their bids because of financing constraints.

8.4 Tariff-jumping and sectoral-growth stories

A depreciation or trade deficit might induce foreign firms to invest in the United States to avoid tariffs or benefit from cheaper U.S. production. These stories can explain more production in the United States, but they do not naturally explain why foreign owners acquire a larger share of domestic assets rather than domestic owners also expanding.

9 Empirical strategy

9.1 Baseline capital-flow regressions

The empirical section estimates regressions of capital inflows into the United States on the real value of the dollar and a time trend. The dependent variables are measured as percent of U.S. GNP. The exchange-rate variable is the real value of the dollar, proxied by the IMF MERM real dollar index, written in the paper as $1/e$. The key sign is negative: a lower dollar value should be associated with higher FDI inflows.

A representative regression is

$$\text{Inflow}_t = a + b(1/e)_t + \gamma t + u_t.$$

The model predicts $b < 0$ for direct investment but not necessarily for passive portfolio flows.

9.2 Disaggregation tests

The paper then disaggregates FDI by industry and by transaction type. This matters because a purely aggregate result could hide a narrow sectoral story. The model predicts broad exchange-rate sensitivity across information-intensive direct investments, not just in one industry.

9.3 Cross-country check

Finally, the paper looks at direct-investment inflows for the United States, United Kingdom, West Germany, Canada, and Japan. These tests are less clean because the Balance-of-Payments data for other countries mix new acquisitions with transfers among existing affiliates, and trade-weighted exchange rates may be poor proxies for the relevant investor wealth shocks.

10 Main empirical results

10.1 FDI differs from passive capital flows

In the quarterly U.S. data, direct investment is the only major private capital-inflow category with a statistically significant negative coefficient on the dollar. Treasury securities have a positive significant coefficient, and portfolio stocks and bonds have a positive but insignificant coefficient. This matches the theory: passive assets have low monitoring costs and should not depend strongly on relative wealth.

10.2 Annual data are similar

Annual regressions give a similar conclusion. The coefficient on direct investment remains negative and significant. The results are not merely high-frequency noise.

10.3 Industry-level evidence

In the industry breakdown, all thirteen exchange-rate coefficients are negative, and five are statistically significant. Manufacturing, especially chemicals, shows strong exchange-rate sensitivity. This suggests that the aggregate U.S. relation is not driven by a single narrow category.

10.4 Transaction-type evidence

The International Trade Administration data show negative coefficients for many transaction types. Mergers and acquisitions, new plants, and joint ventures all have significant negative relations with the dollar for value and/or transaction counts. This is consistent with the model's broad direct-investment channel.

10.5 Other wealth shocks

A key distinction from sectoral-growth stories is that non-exchange-rate wealth changes should also affect FDI. The paper points to Klein and Rosengren's finding that foreign stock-market capitalization relative to U.S. stock-market capitalization predicts U.S. FDI inflows, even after controlling for relative wages. This supports the wealth channel.

11 Extracted figures and tables from the paper

11.1 Figure I: U. S. Foreign Direct Investment Inflows and the Real Exchange Rate; Figure II: Loan Supply Schedule

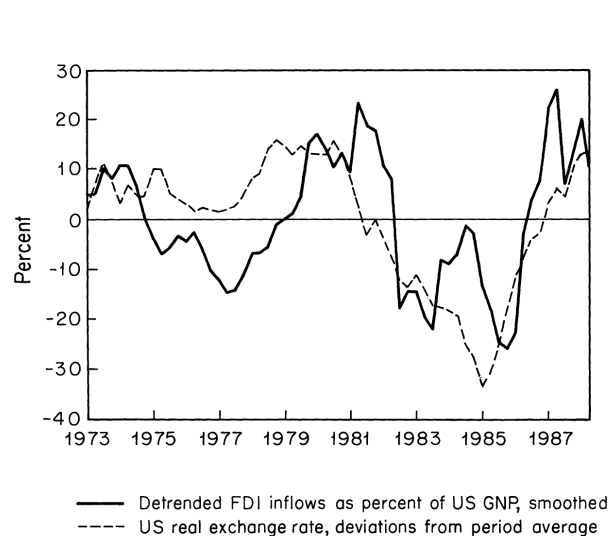


FIGURE I
 U. S. Foreign Direct Investment Inflows and the Real Exchange Rate

Figure I: U. S. Foreign Direct Investment Inflows and the Real Exchange Rate

Interpretation. The visual fact motivates the paper. Detrended FDI inflows move closely with the real exchange rate: when the dollar is weak, FDI inflows tend to be high. The model is designed to explain why this correlation can exist even with internationally mobile capital.

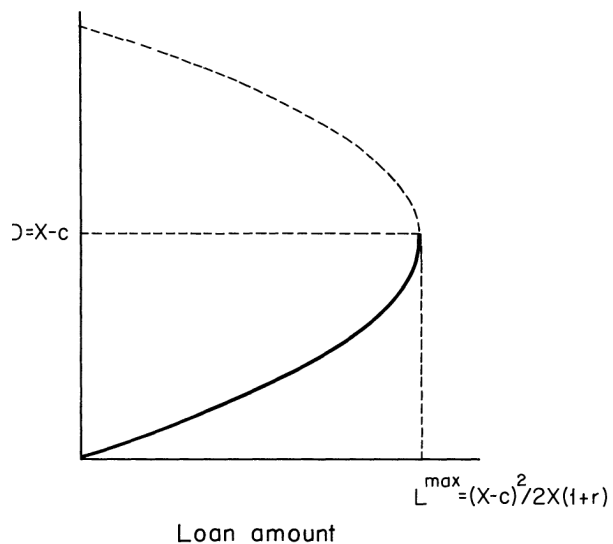


FIGURE II
 Loan Supply Schedule

Figure II: Loan Supply Schedule

Interpretation. The loan schedule shows the financing friction. Because monitoring is costly, increasing the promised repayment eventually lowers expected lender payoff. The maximum loan is below the full expected present value when $c > 0$, which creates credit rationing.

11.2 Figure III: Entrepreneur's Reservation Price as a Function of Wealth;
Figure IV: FDI Inflows as a Function of The Exchange Rate

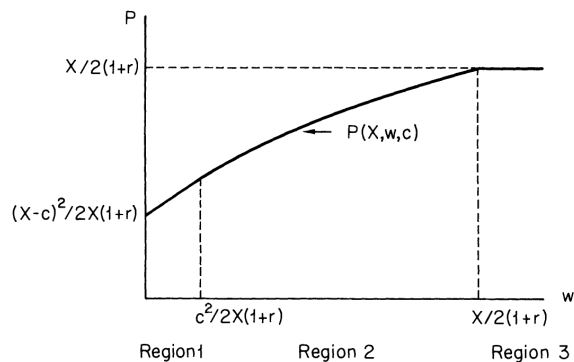


FIGURE III
Entrepreneur's Reservation Price as a Function of Wealth

Figure III: Entrepreneur's Reservation Price as a Function of Wealth

Interpretation. The reservation bid rises with wealth only when external finance is costly. In Region 1 it rises one-for-one with wealth; in Region 2 it rises more slowly; in Region 3 wealth no longer matters. This is the engine behind the exchange-rate channel.

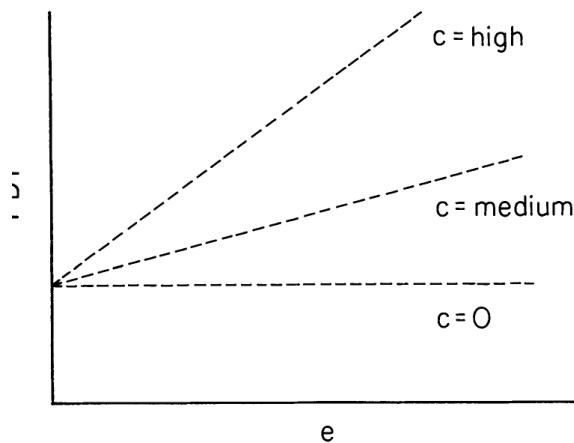


FIGURE IV
FDI Inflows as a Function of The Exchange Rate

Figure IV: FDI Inflows as a Function of The Exchange Rate

Interpretation. The theory predicts that FDI rises with a depreciation, and the slope is steeper for assets with higher monitoring costs. When $c = 0$, exchange rates do not affect FDI because wealth has no effect on reservation prices.

11.3 Table I: Regressions of Changes in Foreign Assets in the United States on the Value of the Dollar (Quarterly Data, 1973–1988); Table II: Regressions of Changes in Foreign Assets in the United States on the Value of the Dollar (Annual Data, 1973–1987)

TABLE I
REGRESSIONS OF CHANGES IN FOREIGN ASSETS IN THE UNITED STATES ON THE
VALUE OF THE DOLLAR (QUARTERLY DATA, 1973–1988)

Form of gross capital inflow into the United States	Coefficients on		DW	R ²	DF
	1/e	t			
Total foreign assets	-0.0962 (0.1501)	0.4141 (0.1020)***	1.79	0.21	59
Foreign official assets	-0.1783 (0.1153)	0.0150 (0.0820)	1.66	0.01	59
Foreign private assets	0.0821 (0.1517)	0.3994 (0.1080)***	1.81	0.32	59
Direct investment	-0.0671 (0.0262)**	0.1336 (0.0190)***	1.92	0.45	59
U. S. Treasury securities	0.0943 (0.0330)***	0.0090 (0.0250)	1.63	0.19	59
Corporate stocks and bonds and other bonds	0.0575 (0.0541)	0.1580 (0.0460)***	0.59	0.31	59

Note. Standard errors in parentheses are the larger of those estimated allowing for both autocorrelation and (i) homoskedasticity and (ii) heteroskedasticity. *, **, *** represent statistical significance at the 10, 5, and 1 percent levels, respectively. Regressions were estimated with constant terms, which are not reported. Legend: 1/e—index of IMF merm real value of the dollar (1980 = 100); t—time trend. Dependent variable is expressed in percent of U. S. GNP.

TABLE II
REGRESSIONS OF CHANGES IN FOREIGN ASSETS IN THE UNITED STATES ON THE
VALUE OF THE DOLLAR (ANNUAL DATA, 1973–1987)

Form of gross capital inflow into the United States	Coefficients on		DW	R ²	DF
	1/e	t			
Total foreign assets	-0.0619 (0.1816)	1.6460 (0.5170)***	1.57	0.45	12
Foreign official assets	-0.0117 (0.0114)	0.0060 (0.0320)	1.54	0.00	12
Foreign private assets	0.0798 (0.1236)	1.7274 (0.3520)***	1.94	0.74	12
Direct investment	-0.0832 (0.0339)**	0.5685 (0.0960)***	2.13	0.71	12
U. S. Treasury securities	0.1331 (0.0371)***	-0.0870 (0.1000)	1.22	0.55	12
Corporate stocks and bonds and other bonds	0.0289 (0.0965)	0.7280 (0.2750)	0.97	0.41	12

Note. Standard errors in parentheses are the larger of those estimated allowing for both autocorrelation and (i) homoskedasticity and (ii) heteroskedasticity. *, **, *** represent statistical significance at the 10, 5, and 1 percent levels, respectively. Regressions were estimated with constant terms, which are not reported. Legend: 1/e—index of IMF merm real value of the dollar (1980 = 100); t—time trend. Dependent variable is expressed in percent of U. S. GNP.

Table I: Regressions of Changes in Foreign Assets in the United States on the Value of the Dollar (Quarterly Data, 1973–1988)

Interpretation. The key coefficient is direct investment: -0.0671, significant at the 5 percent level. Passive categories do not show the same pattern. Treasury securities have the opposite sign, which is consistent with the theory's distinction between direct control assets and passive portfolio assets.

Table II: Regressions of Changes in Foreign Assets in the United States on the Value of the Dollar (Annual Data, 1973–1987)

Interpretation. The annual data reproduce the quarterly pattern. Direct investment has a negative significant coefficient, -0.0832, while portfolio and Treasury categories do not behave like FDI. This weakens the concern that the result is only high-frequency noise.

11.4 Table III: Regressions of FDI Inflows by Industry (Balance of Payments Data, Annual 1974–1987); Table IV: Regressions of Foreign Direct Investments Purchased (International Trade Administration Data, Annual 1977–1987)

TABLE III
REGRESSIONS OF FDI INFLOWS BY INDUSTRY (BALANCE OF PAYMENTS DATA,
ANNUAL 1974–1987)

Industry	Coefficients on		DW	R ²	DF
	1/e	t			
All industries	−0.0876 (0.0320)**	0.6872 (0.1219)***	1.88	0.70	11
Petroleum	−0.0020 (0.0112)	0.0626 (0.0431)	2.46	0.05	11
Manufacturing	−0.0516 (0.0160)***	0.2792 (0.0604)***	1.81	0.61	11
Food	−0.0064 (0.0044)	0.0553 (0.0173)***	2.90	0.39	11
Chemicals	−0.0216 (0.0068)***	0.1156 (0.0263)***	2.10	0.58	11
Fabricated metals	−0.0016 (0.0032)	0.0082 (0.0120)	2.69	0.00	11
Machinery	−0.0144 (0.0052)**	0.0497 (0.0193)**	1.54	0.37	11
Other manufacturing	−0.0088 (0.0040)**	0.0600 (0.0147)***	1.78	0.53	11
Trade	−0.0020 (0.0060)	0.0644 (0.0224)**	2.03	0.40	11
Finance	−0.0088 (0.0088)	0.0643 (0.0337)*	3.11	0.11	11
Insurance	−0.0032 (0.0064)	0.0389 (0.0239)	1.72	0.07	11
Real estate	−0.0048 (0.0060)	0.0641 (0.0229)**	2.11	0.34	11
Other industries	−0.0128 (0.0080)	0.0927 (0.0301)***	2.17	0.37	11

Note. Standard errors in parentheses are the larger of those estimated allowing for both autocorrelation and (i) homoskedasticity and (ii) heteroskedasticity. *, **, *** represent statistical significance at the 10, 5, and 1 percent levels, respectively. Regressions were estimated with constant terms, which are not reported. Legend: 1/e—index of IMF term real value of the dollar (1980 = 100); t—time trend. Dependent variable is in percent of the U. S. GNP.

Table III: Regressions of FDI Inflows by Industry (Balance of Payments Data, Annual 1974–1987)

Interpretation. All industry coefficients on the dollar are negative, and several are statistically significant. Manufacturing, chemicals, machinery, and other manufacturing show strong sensitivity. This supports a broad FDI effect rather than a single-sector result.

TABLE IV
REGRESSIONS OF FOREIGN DIRECT INVESTMENTS PURCHASED (INTERNATIONAL
TRADE ADMINISTRATION DATA, ANNUAL 1977–1987)

Type of investment	Coefficients on		DW	R ²	DF
		1/e t			
All types:	value	−0.1094 (0.0344)***	1.1135 (0.1813)***	2.70	0.78 8
	number	−4.9442 (1.6214)***	38.2377 (8.5642)***	1.05	0.65 8
Mergers and acquisitions:	value	−0.057 (0.0168)***	0.5621 (0.0893)***	2.14	0.79 8
	number	−1.8996 (0.5068)***	12.62895 (2.6766)***	2.35	0.83 8
Equity increases:	value	−0.0092 (0.0064)	0.1277 (0.0341)	2.39	0.56 8
	number	−0.1224 (0.0462)**	1.4494 (0.2437)***	2.39	0.77 8
Real estate:	value	−0.009 (0.0184)	0.04025 (0.0968)	1.00	0.00 8
	number	−0.8512 (1.0958)	−0.3084 (5.7884)	0.64	0.00 8
New plant:	value	−0.0098 (0.0028)***	0.0817 (0.0143)***	2.58	0.76 8
	number	−0.8124 (0.2180)***	5.32375 (1.1514)***	1.81	0.68 8
Joint ventures:	value	−0.0056 (0.0024)**	0.0651 (0.0126)***	2.28	0.72 8
	number	−0.2422 (0.1002)**	3.8512 (0.5291)***	2.27	0.82 8
Plant expansion:	value	−0.004 (0.0010)***	0.046 (0.0051)***	2.07	0.89 8
	number	−0.2186 (0.1666)	2.6582 (0.8796)***	1.80	0.42 8
Other expansion:	value	−0.0066 (0.0056)	0.221 (0.0304)***	2.39	0.86 8
	number	−0.052 (0.3050)	9.4884 (1.6113)***	2.52	0.82 8
No type listed:	value	−0.0078 (0.0088)	−0.03035 (0.0468)	2.24	0.06 8
	number	−0.7458 (0.8630)	−3.03925 (4.5580)	2.24	0.06 8

Note. Standard errors in parentheses are the larger of those estimated allowing for both autocorrelation and (i) homoskedasticity and (ii) heteroskedasticity. *, **, *** represent statistical significance at the 10, 5, and 1 percent levels, respectively. Regressions were estimated with constant terms, which are not reported. Legend: 1/e—index of IMF term real value of the dollar (1980 = 100); t—time trend. Dependent variable is in percent of U. S. GNP.

Table IV: Regressions of Foreign Direct Investments Purchased (International Trade Administration Data, Annual 1977–1987)

Interpretation. The ITA transaction data show that the dollar relation appears in transaction counts and values. Mergers and acquisitions, new plants, joint ventures, and plant expansions are especially informative because these are direct-control transactions rather than passive purchases.

11.5 Table V: Country Regressions of Direct Investment Inflows on Exchange Rates (Annual Data, 1972–1987)

TABLE V
COUNTRY REGRESSIONS OF DIRECT INVESTMENT INFLOWS ON EXCHANGE RATES
(ANNUAL DATA, 1972–1987)

Country	Coefficients on		DW	R^2	DF
	$1/e$	t			
United States	−0.0948 (0.0357)**	0.6564 (0.0969)***	2.14	0.75	13
United Kingdom	−0.0050 (0.0165)		1.57	0.00	14
West Germany	−0.0103 (0.0030)***		1.22	0.52	14
Canada	0.0084 (0.0287)		1.52	0.00	14
Japan	−0.0006 (0.0027)		2.40	0.00	14

Note. Standard errors in parentheses are the larger of those allowing for both autocorrelation and (i) homoskedasticity and (ii) heteroskedasticity. *, **, *** represent statistical significance at the 10, 5, and 1 percent levels, respectively. Regressions were estimated with constant terms, which are not reported. Legend: $1/e$ —index of IMF term real value of the dollar (1980 = 100); t —time trend. Dependent variables are country FDI inflows in percent of country GNP.

Table V: Country Regressions of Direct Investment Inflows on Exchange Rates (Annual Data, 1972–1987)

Interpretation. The cross-country check is weaker than the U.S. evidence. The United States and West Germany show significant negative coefficients; Canada has the wrong sign using a multilateral exchange rate, though the paper notes that a bilateral U.S.-Canada rate makes the coefficient negative. Data limitations make this table suggestive rather than decisive.

12 How the results fit together

12.1 Theory-to-data link

The model predicts that exchange-rate movements should matter most where internal wealth matters most. Direct investments require control and involve more information asymmetry, so they depend on investor wealth. Passive investments are easier to finance externally, so they should not respond in the same way.

The tables line up with this logic: FDI responds negatively to the value of the dollar, while Treasury and portfolio inflows do not. The fact that the result appears across industries and transaction types strengthens the interpretation.

12.2 Why the exchange-rate coefficient is negative

The coefficient on the real value of the dollar is negative because a stronger dollar reduces the domestic-currency wealth of foreign investors, while a weaker dollar raises it. When the dollar depreciates, foreign investors can bid more for U.S. direct assets, increasing FDI inflows.

12.3 Why passive capital flows are the comparison group

Passive capital flows provide a built-in falsification check. If all capital inflows mechanically rose when the dollar fell, then Treasuries and portfolio securities should show the same pattern as FDI. They do not. This difference is the central empirical support for the imperfect-capital-markets interpretation.

13 Limitations and caveats

13.1 Small samples and time trends

The empirical samples are short, especially in annual and cross-country regressions. Time trends are included because the United States became increasingly important as a host for foreign investment. The results are strong enough to be suggestive, but not designed as a modern high-dimensional causal design.

13.2 Measurement of relevant wealth

The real exchange rate is a proxy for relative wealth. It is imperfect because foreign investors differ by country, asset currency, and portfolio composition. Other wealth measures, such as relative foreign stock-market capitalization, are useful complements.

13.3 FDI data definitions

Balance-of-Payments definitions classify direct investment using ownership thresholds. Some recorded FDI may include intra-company transfers rather than new acquisitions or new plant formation. This is why the ITA transaction-type evidence is important.

13.4 Welfare cannot be inferred from flows alone

Even if depreciation increases foreign acquisitions, welfare depends on asset values, auction competition, financing constraints, and the productivity of foreign control. The flow result does not automatically imply that foreign acquisitions are harmful.

14 Conclusion

Froot and Stein provide a clean mechanism linking exchange rates to FDI through relative wealth and imperfect capital markets. The model preserves global capital mobility for riskless borrowing but adds informational frictions in financing direct-control assets. This combination lets exchange rates affect the composition of capital inflows.

The paper's most important insight is that FDI is not just another form of capital inflow. Direct investment is information-intensive and requires internal funds, so currency-induced wealth shocks affect who can buy assets. The empirical evidence supports this distinction: FDI rises when the dollar weakens, but passive portfolio categories do not show the same pattern.